DISCRETE MATH · SET THEORY

Venn Diagram & Membership Table Explorer Explore & Reflect

Name _____ Date _____ Period _____

Open **Venn Diagram & Membership Table Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. $U = \{1, \ldots, 20\}$, A = evens, B = multiples of 3. Predict $A \cap B$ and $(A \cup B)'$.

2. Is $(A \cup B)'$ the same set as $A' \cup B'$? Predict, then check.

2 Explore use the tool

1. Shade $A \cap B'$ yourself, then type it as Expression 1. List its elements.

2. Make Expression 1 = $(A \cup B)'$. Find an Expression 2 with $\cap$ and no $\cup$ that is equal. Which rows of the membership table did you check?

3. Write A $\oplus$ B without $\oplus$ and list its elements.

4. Three sets (C = primes): shade the elements in exactly one of A, B, C. List them.

3 Record use the tool and fill in the table

Membership table: is it an identity?

Three sets. Type both sides; record whether every row agrees.

Left side	Right side	Equal?	A region where they differ (if any)
$A - (B \cup C)$	$(A - B) \cap (A - C)$		
$A \cup (B \cap C)$	$(A \cup B) \cap (A \cup C)$		
$A - (B - C)$	$(A - B) - C$		
$(A \cap B)'$	$A' \cup B'$		

4 Explain what did you discover?

1. The complement X' contains everything in _____ that is not in X.

2. De Morgan: $(A \cup B)' =$ _____ and $(A \cap B)' =$ _____.

3. With n sets there are _____ regions, so a membership table has 2^n rows.

4. Two expressions are equal sets for all A, B, C exactly when they contain the same _____.

5. Show with an element that $A - (B - C) \neq (A - B) - C$ for these sets, and explain why one element is enough.

5 Apply a new situation, no tool

1. $U = \{1, \ldots, 10\}$, $P = \{2, 3, 5, 7\}$, $Q = \{1, 3, 5, 7, 9\}$. Find $P \cap Q'$ and $(P \cup Q)'$.

2. $n(A) = 40$, $n(B) = 25$, $n(A \cap B) = 10$. Find $n(A \cup B)$ and $n(A \oplus B)$.

3. Simplify $(A \cap B) \cup (A \cap B')$.

4. Draw a Venn diagram shading $(A \cup C) \cap B'$.